DOG MATH

Written by Claude (Anthropic), 2026-10-20 15:58.

A dog: a doc written in dog math, about dog math. The rules are doc's, from the ledger and from chat; the words are the model's. Not doc's text.

1. Playing with ideas

Dog math is about playing with ideas. When hands, cats, lumps, thumps, pow and zoom are on the board, the work is real science or real engineering, and it says out loud that it isn't the last word. It won't get lost in the weeds, because the weeds were rounded off at the gate.

Dog math is the default for all the work, unless doc says a piece is for outside eyes (sections 14 and 16).

The job is naming. Find the quantity that means something, say how to measure it or build it, and relate it to the others in sentences that don't turn round on themselves. A lump of algebra, or a sentence that tangles or runs long, means a quantity hasn't been named yet: find the body or setup property it's reaching for, name it, and let it eat the algebra. The quantity does the structural work, and friendly units make the numbers easy.

2. House rules

1. Round hard. Gravity is 10 zoom, dog gravity. Anything close to a round number is that number; a mole is 3/5 of a trillion trillion. A third digit is margin noise: 3.92 is 3.9, and 0.533 is about 0.53.

2. A result survives rounding, dropping, even scrambling, every digit past about the third. If 96 against 100 matters, you're measuring the wrong thing: pick a simpler example, or find the real quantity.

3. Pick sizes so the units eat the algebra. Radii in whole hands make a whole number of cards, and where algebra seems to be needed a unit on the board usually eats it. Pick values that aren't halfway: a half hides which part is which.

4. Local units are fine. A problem can define its own at the top

when it makes the numbers whole: 1 mw is a milliwheel, so a lump at a hand, at a hand, is 10 mw.

5. Every quantity gets a one-word unit: its own name, unless a named unit exists already. After a number you say "wheel", not "inertia". Force is both quantity and unit (five force, and the dog unit is the push); a new quantity gets the same deal.

6. Dog units take no plural s: 5.3 hand, 7 lump, 14 push. 5.3 hands is a trip to the hospital.

7. Words mean what they mean. A name holds up against its dictionary meaning everywhere you can see it mapped; one name per thing, one thing per name. Assigning a word never bans its everyday sense: "spinning at 6" is still the sweep in plain English, so spin stays out of its way. Spin the quantity is the turning movement, inertia at sweep; the rate is the sweep. Where both are in play, say sweep or turning movement and leave the bare word out. Each word is a unit test on the algebra.

8. The verbs are fixed by the old rules of the language. Energy is transformed. Work is performed. Punch and hook are delivered, or land. A spring returns what was put into it. Fuel is burned. Motion is traded, or added over time. A force does work along a path. Effort is taken, like a load: holding a load up takes effort and does no work. Spending is for energy, power and motion.

9. Make the words work hard. A quantity can go adjectival (a ponderous bulk), adverbial, or get verbed (the drum picks up). Drop dead words, skip words, take shortenings, have fun with units, and don't police anyone's coinages. Formalize later, and only where it would step on toes.

10. Operators first. Assume crane and bulldozer operators understand this better than physicists, and adopt their language whenever possible.

11. Balanced statements, always. One-sided versions are for engineering.

12. What is simpler in turns is written in turns.

13. A letter means one thing in a problem, and never a unit's. Don't spend m on a phase with a metre on the page, or s on a swing beside seconds, or C on a charge beside a can. Give

a phase φ and a swing θ, and leave the unit its letter.

14. Two observers are A and B: A is the active agent, the one who flashes and counts, and B is the subject, the one being watched. "A's own time" and "B's own time" say whose clock without a new letter, and "one's own" does it for anyone. With more than two, give names or numbers. (Not me and you: those are for the writer and the reader.)

15. Dog math says whatever the usual words say. It is those words plus relabelings, so it is never allowed to be less clear than they are. Where a usual name is clearer in the scene, say it, plainly and without comment: topspin and backspin, simple harmonic motion, the parallel-axis theorem, time dilation, angular momentum, event, frame. A name for a whole result is a handle, and a handle is just English. And where a dog word collides with something in the scene, the scene keeps its word and the dog word steps aside.

Where a dog word steps aside

Each of these is a rock on the trail. Point at it in passing, as if obvious, and move on.

- roll. In a problem with something rolling, the spin is the turning movement (or angular momentum), and its number goes in kg m^2/s. Roll stays a unit where nothing rolls.

- wheel. Not as a unit in a problem about an actual wheel, nor for a rod or a pendulum. Use mw, or kg m^2.

- forward and back around. Where the scene has a word for the sense of turning, topspin, backspin, the way a wheel rolls, use it (section 9).

- outwardness and upwardness. They name parts along and across a start line, wherever it points. Say which way when it isn't up and out (section 9).

- across. Only the product: a mass across a profile. A direction is sideways, level, or at right angles to.

- run. Only a speed: it runs at 3 cat. A computer's work is a check, a whole motion is a motion.

- frame, event. A frame is how places and times are counted. An event is a place at a time. Surface and floor are things you

stand on.

- moment. It has three jobs and the dog words keep them apart: the instant, the lever, the inertia.

- punch, in a problem about a blow. The strike is a blow or a hit; punch is what it leaves in the centre's motion.

- heft. Weight is the pull, heft is what gravity stores. The weight pulls; the height hefts.

- swing. The angle swung through. Its biggest is the amplitude, and one trip out and back is a cycle.

- can, in a problem with a can on the page. Ask "could", not "can": could the punch rise while the pace stays put?

- coil. The coil is the wire and the store it makes. With a second coil in the story, such as the wire on a swinging rod, say which: the wire on the rod, the coil's store.

3. The units

The dog units:

Unit	What it is	Standard
hand	about a hand's width	10 cm (~4 in)
card	a square hand; about a 3x5 index card	0.01 m^2
sheet	a square metre of area	1 m^2
blanket	ten sheet	10 m^2
lump	the unit of mass	1 kg
cat	the unit of speed	1 m/s
dog	ten cats	10 m/s
zoom	a cat per second	1 m/s^2
dog gravity	a dog per second, down	10 zoom
push	force: an oompf per second	1 N
pull	a push, pulling	1 N
press	pressure: a push per sheet (Pa for company)	1 Pa
thump	a lump lifted a hand in dog gravity	1 J
pow	a thump per second	1 W
crank	torque: a push at a metre, an arrow along the axis	1 N m

grunt	trundle: the work over one stroke, a plain amount	1 J, one stroke's worth

The book's own units are the one-word units of the quantities the book assigns. Oompf, roll, ponder and wheel stay, for outside eyes too, wherever the quantities are named:

Unit	Quantity it measures	Standard
oompf	movement, a body's motion	1 kg m/s
stroke	swing (one radian's worth)	1 rad
quadrant	swing: a quarter turn	about 1.6 stroke
flip	swing: half a turn	about 3.1 stroke
turn	swing: all the way round	about 6.3 stroke
sweep	sweep: a stroke per second	1 rad/s
pickup	pickup: a sweep per second	1 rad/s^2
whirl	pickup of a body turning on itself: a sweep per second. A train picking up around a track's centre counts in pickup, not whirl	1 rad/s^2
ponder	bulk: mass at a lever ("a ponderous bulk")	1 kg m
wheel	inertia: a ponder at a metre	1 kg m^2
roll	spin: turning movement	1 kg m^2/s
shift	shift: movement for a time; what moves a centre of mass	1 kg m
transit	transit: the swing of the bulk, mass along the arc	1 kg m
zip	punch per mass	1 J/kg
zing	hook per inertia	1 J/(kg m^2)
zump	hook per mass: profile across zing	1 J/kg
zug	work per mass, or per inertia| for turning	1 J/kg, or 1 J/(kg m^2)
zow	power per mass, or per inertia for turning	1 W/kg, or 1 W/(kg m^2)
zam	action per mass, or per inertia for turning	1 J s/kg, or 1 J s/(kg m^2)
zlug	work per inertia, when both kinds are about at once	1 J/(kg m^2)
zlow	power per inertia, likewise	1 W/(kg m^2)
zlam	action per inertia, likewise	1 J s/(kg m^2)
swirl	spin per mass; for a point, the lever at pace; for a body, lever each bit's	1 m^2/s

	movement, add, then divide	
	by the mass; along the axis	
twirl	torque per mass; for a point, the lever at acceleration; for a body, lever each bit's push, add, then divide by the mass (a balanced wheel can twirl with its centre still); along the axis	1 m^2/s^2

Ring is the phase rate of anything that oscillates or waves, a stroke of phase a second (trial).

Sheet and blanket count the area of something laid out: a wing, a field, a floor. Card counts profiles, which carry no mass. The shapes are alike and the jobs differ.

Handy facts:

1 metre	= 10 hand
1 sheet	= 100 card
1 blanket	= 10 sheet
1 wheel	= a ponder at a metre
a lump at a hand, at a hand	= 10 mw (local unit)
1 thump	= a lump lifted a hand, dog gravity
1 crank through 1 stroke	= 1 grunt
dog gravity	= 10 zoom
a turn	= about 6.3 stroke
a 3x5 card	= within a few percent of a card

4. Saying products

Products and quotients take English operators, not ×, and get verbed when the verbing's good.

of	leads with the operator: the sweep of the lot, the capacitance of its voltage
at	leads with the thing: the lot at pace, a mass at a lever
across	a mass across a profile; a wheel across a zing
through	a push through a distance
on	divides: 3 roll on half a wheel
over	divides, or runs along: 4 pickup over 4.5 stroke
under	a bulk under gravitation

"Of" is the ordinary genitive, not a fixed operator. The product

is whichever the two kinds of object take: two axis arrows dot, an axis arrow and a lever cross, a scale just scales. Two orders, two words: "of" leads with the operator, "at" leads with the thing. Running text uses the sentence that comes out positive.

Where "of" could be heard as belonging instead of times, and the two hearings lead somewhere different ("the power is the voltage of the current"), put the numbers beside it the first time: 12 V of 2 A is 24 W. After that the bare rule is safe.

Said	Is
1 hand sweeping at 50	pace, 5 cat
7 lump moving at 7 cat	movement, 49 oompf
2 oompf levered at 5 hand	spin, 1 roll
0.5 wheel sweeping at 2	spin, 1 roll
1.5 crank sweeping at 2	power, 3 pow
2 lump at 5 hand	bulk, 1 ponder
18 zing of half a wheel	hook, 9 thump
2 lump across 2 card	inertia, 40 mw
4 push through 3 metre	work, 12 thump
0.1 ponder under 10 zoom	heft, 1 thump

Duration and rate are rows in the multiplication table, not columns. Say "z is the fluxion of y" (force is the fluxion of the movement) and "y adds z over time" (a push adds movement over time). Mass is handled by the ghost version beneath each cell.

5. Verbing

A unit worth introducing has a verb in it, or if not, an adverb or an adjective. The centre of verbing is what's happening.

- A push pushes, and a pull pulls: 4 pull on 2 lump, steadily pulling at 4 along the ice. A crank cranks. A boost boosts. A wheel wheels: 0.27 wheel wheeling at 5.0 carries 1.35 roll. A body hooks, punches and hefts.

- Movement and spin are carried: the cart carries 6 oompf, the wheel 3 roll. The ball gives the bat oompf, the blow boosts it, and the bat rolls with it.

- Ponder is from the Latin for "to weigh". You ponder from a pivot: the rod ponders from P, and its bulk under gravity cranks it back. A ponderous bulk.

- Whirl is the pickup of a body turning on itself, so it belongs where the body is in the sentence: 0.8 crank on 40 mw picks the puck up at 20 whirl. A train picking up round a track's centre picks up. A sentence with no body in it picks up too: 20 pickup over 10 stroke adds 200 zing.

6. Ghosts

A ghost body is a body with the mass taken out. A ghost still needs something: a place to be, or some other property to have. Some ghosts don't need to be anyplace in particular; they're sneaky that way, though bodies get the same latitude (2 lump at 3 cat doesn't say where). One can have only a velocity and a swirl. We add places when there's more than one thing to keep apart. What a ghost can't be is no matter, no place and no anything at all. That's the ghost of a ghost, and it's just gone.

The centre of mass is the plainest ghost body: a place, a velocity, a swirl about any line, and no mass of its own. A laser dot is another.

A ghost quantity is a quantity with the body taken out. Velocity, swirl and zip are a ghost's properties, the way movement, spin and punch are a body's. Straight ghosts come per lump; turning ghosts come per wheel. Some turning quantities also have a per-lump ghost.

	per lump (mass out)	per wheel (inertia out)
oompf	velocity	
push	acceleration	
thump, punch	zip	
wheel	profile	
roll	swirl	sweep
crank	twirl	pickup
thump, hook	zump	zing
thump, work	zug	zlug
pow	zow	zlow

Take the wheel out of a 2-lump ball at 5 hand, which is half a wheel:

    3 roll on half a wheel sweeps at 6.

2 crank on half a wheel picks up at 4.
9 thump of hook on half a wheel is 18 zing.
4 pickup over 4.5 stroke is 18 zlug.
4 pickup at 6 sweep is 24 zlow.

Take the lump out of the same ball instead:

half a wheel on 2 lump is 25 card of profile.
9 thump of hook on 2 lump is 4.5 zump.
25 card of profile across 18 zing is 4.5 zump.

Spin and torque per wheel leave the motion itself, sweep and pickup, just as movement and push per lump leave the velocity and the acceleration. Spin on the wheel gives the sweep wherever the setup fixes the axis. A crank on the wheel gives the pickup only while the inertia stays put: a skater tucking on a fixed axis picks up with no crank at all.

Some ghosts we're so used to we've stopped seeing them: velocity, speed, acceleration. One way to sort them: real quantities add, ghosts average, each by its own weight, the mass for the per-lump ghosts and the inertia for the per-wheel ones. Glue a 2-lump ball at 3 cat to a 1-lump ball at 6 cat, both moving the same way:

movement: 6 oompf and 6 oompf make 12 oompf. It adds.
velocity: 3 cat and 6 cat make 4 cat, the 12 oompf spread
        across 3 lump. It averages, by the mass.

That's bookkeeping, not a rule about what an interaction changes. Gluing is a collision. Movement comes through it, so the velocity comes out as the mass-weighted average. Punch doesn't all come through:

before: 27 thump of punch, 9 zip
after: 24 thump at 4 cat, 8 zip; 3 thump burned

Movement, spin, punch and hook are carried by bodies, and can be split up and swapped around between them. Velocity, speed, zip and acceleration belong to no body in particular, and split only by the mass. Ghosts aren't weak for that. A free body keeps its velocity, and velocities say where bodies will meet. What they can't settle alone is how much motion gets traded when bodies interact: that takes the masses and the rules of the interaction.

Dog gravity is a ghost: 10 pull a lump, 10 zoom. The mass is already out, so everything falls together.

Ghosts carry no mass, so they walk through walls, and sometimes

they let you follow them if you dare. People get lost in those walls. They say if you could see into the dark in there, it would be nothing but algebra, everywhere; be grateful the lights are off. The way out is always the same: put the body back across. You go in after 18 zing, and on the far side half a wheel steps out and hooks 9 thump. The ones who get lost forgot to bring the mass back.

7. Verbing the ghosts

A ghost doesn't have to sit after a number. Let it run. Put a body, or a slice of one, across a ghost, and it takes the ghost's verb:

    5 hand sweeping at 6 runs at 3 cat.
    Half a wheel sweeping at 6 hooks 9 thump.
    Half a wheel zinging at 18 hooks 9 thump.
    2 lump zipping at 2 punches 4 thump.
    1 lump dropping a hand hefts 1 thump.

Per lump, the profile takes the same verbs: 25 card sweeping at 6 hooks 4.5 thump a lump.

Ghost	Its verb	Said
velocity	runs at	the centre runs at 3 cat
acceleration	speeds up at	the block speeds up at 1.8 zoom
sweep	sweeps at	5 hand sweeping at 6
pickup	picks up at	the drum picks up at 18
zip	zips at	2 lump zipping at 2
zing	zings at	half a wheel zinging at 18
swirl	swirls	the ball swirls the same after the bounce as before
twirl	twirls	a balanced wheel can twirl with its centre still

Ghosts take the passive voice too. The point is to pass them around without anything in particular having them: 18 zing is added, not "it gains 18 zing". Or let the quantity act: 4 pickup over 4.5 stroke adds 18 zing. There's still a ghost body back there, but it's rude to point at it. Put a body back across a ghost and the active voice comes back with it: half a wheel with 18 zing hooks 9 thump.

Don't force the verbing. Where a verb's word hasn't come in yet,

the one that has will carry it: before zing is on the table,
"sweeping at" carries the hook. And round as you verb.

8. Ghost manners, and the letter

Mention the ghosts enough that nobody is confused, then leave them
be. Most people are afraid of ghosts, so hold the reader's hand
and make no sudden moves.

- Don't point. No possessives: "the rod's profile" becomes "the
  profile about the centre". No roll call at the end of each
  problem. No "here's another ghost": name the quantity and let
  it work. "Spin per mass we'll call the swirl."

- Say what a ghost is once, plainly, where the reader first
  needs it: a quantity with the body's mass, or its inertia,
  taken out. After that the ghosts do their work quietly.

- The familiar ones, velocity and the sweep, soothe, and need
  no letter. A problem with one scary ghost can give it a
  letter. That letter is Ω, the ghosts' true form, the base
  letter for the whole family: Ω for the one scary ghost, and
  Ω_profile, Ω_zing, Ω_zlow when several are in play. Marks
  stay (Ω_{1}, Ω_{2}, Ω_{+}, Ω_{-}); a different ghost gets a different
  subscript, and a repeated symbol is the same quantity.

- The conventional quantities keep their usual symbols: I, K,
  ω, α, v and, in electricity, Z for impedance. The ghosts give
  way to them, never the other way round. Ω the ghost never
  shares an expression with the ohm: quantity algebra and unit
  statements stay separate.

- All the ghosts wear Ω, or none do. Nobody gets left visible
  alone. If a later room needs the switch, the earlier rooms
  change with it.

9. Turning words

A stroke is a radian. A turn is about 6.3 stroke, and what is
simpler in turns is written in turns. The quadrant and the flip
are a quarter and a half of it.

The two senses of turning are forward around (the default) and
backward around. Say them in full on the first mention in each
problem, like each new newspaper story. After that forward around
shortens to "around"; backward around never does, but "back

around" is fine. Widdershins and sunwise can be mentioned once, where they are defined.

Where the scene has its own word for the sense, use it. Forward around is the way a point goes widdershins; it is not the way a ball travels. A ball rolling to the right turns back around, which is topspin, and a blow below its centre starts it turning forward around, which is backspin.

Arrows get their parts named, in plain composition. A 5 cat run up a 3-4-5 slope has an upwardness of 3 cat and an outwardness of 4 cat. Divide the arrow by up and the 3 comes out; by out, the 4. Say upwardness, how much of the way up, fall, lift. Work with the parts. When the swing is the variable, the parts have names too: the upwardness of a swing and the outwardness of a swing are the parts of an arrow one long, swung forward around from straight out. A 5 cat run swung 0.64 stroke has 5 upwardness of 0.64, 3 cat up, and 5 outwardness of 0.64, 4 cat out. So a bead R from the centre, swung s from straight out, sits R outwardness of s across and R upwardness of s up.

The parts belong to the start line, not the compass. Outwardness is the part along the start line and upwardness the part across it, wherever it points. A bead swung s from straight down sits R outwardness of s down and R upwardness of s sideways: say "down" and "sideways" with them. A wave has no start line; its outwardness and upwardness of the phase are the two shapes a quarter turn apart.

A body turning has a sweep. A swing or a wave also has a ring, a stroke of phase a second: a pendulum sweeps when it moves, and rings at its own rate when it swings.

A swing at one ring and no other is a tone (on trial): the tone rings at its ring. It is what is called simple harmonic motion: a pickup of minus the ring, squared, across the swing. Use either name; the old one is a handle for the whole result. A system with two ways to ring on its own has two tones, each with its own ring; both at once is a chord, and the chord beats at the gap between its rings. The two tones of a rod coupled to a loop, say, ring at 1.6 and 3.6, and beat at 2.0.

10. Balanced, with the d's on different sides

Write the balanced version: both sides of every push, both ledgers, stated positive direction.

Put the differentials on different sides of the balance. They can divide out either way, they're not picky, and then you're not futzing with fractions. Read both layouts aloud, and use the one that tells the physical transfer: what acts, what moves through what, and what is stored.

4 pull on a 2-lump block on smooth ice, for 2 seconds:

F dt = dp 4 pull across 2 s adds 8 oompf
2 lump with 8 oompf runs at 4 cat

F dx = d(½ m v²) 4 pull through 4 metre adds 16 thump of punch: 2 lump at 4 cat punches 16 thump

The rate form, dK/dt = F v, throws away most of the information and compresses the answer. The d's apart say who pushes, how far it moves, and what is kept.

Exchange rates (on trial)

When the change is more than a plain rate, or both sides are crowded, usually (not always) break it out. Ask what is really being carried, from what to where, and at what exchange rate. A store changes by the rate across the change in what is carried, one term for each thing carried:

carried	rate	the store changes by
oompf	pace	pace across d(oompf)
roll	sweep	sweep across d(roll)
metre	pull	pull across d(metre)
stroke	crank	crank across d(stroke)
charge	voltage	voltage across d(charge)
linked flux	current	current across d(flux)

The rate is thump per unit carried: a pace is thump per oompf, 1 cat, so the units check themselves. A plain rate stays a plain rate; break out only when the plain version hides the exchange.

Each body has its own rate. A 2 lump at 3 cat meets a 1 lump at rest and they stick. The 2 lump hands over 2 oompf, at paces running from 3 down to 2 cat: it loses 5 thump. The 1 lump takes the same 2 oompf at paces from 0 up to 2: it gains 2 thump. The 3 thump between is the burn, the gap between the two paces summed over the oompf swapped.

When the carrying changes the rate itself, there is a second term for it. Roll kept and the wheel shrinking: the hook rises by half

a zing across the wheel lost. That is the skater drawing her arms in; the hook she gains is what her arms spent.

11. Electricity

Current is the conductance of the voltage: $I = G V$
Voltage is the resistance of the current: $V = R I$
Power is the voltage of the current: $P = V I$

A resistor burns power, turning electrical energy into heat. Current is charge's fluxion; voltage is energy per charge. "Of" supplies the multiplication; "fluxion" supplies the rate of change.

Can and coil are good strong words. They work as nouns, verbs, gerunds, all kinds of things. A can cans charge; a coil coils flux.

The can cans the separated charge, Q.
The charge a can cans is the capacitance of its voltage:

$$Q = C V$$

The coil coils the linked flux, λ. A coil is a helix: the wire goes N turns around the axis, stepping along it as it goes, so the flux through one turn counts N times. The flux a coil coils is the inductance of its current:

$$\lambda = N \Phi = L I$$

The coil gets the rule that mirrors the can's. Current is charge's fluxion; voltage is linked flux's fluxion:

$$I = dQ/dt, \quad V = d\lambda/dt$$

A can stores the voltage of added charge. A coil stores the current of added flux. Same move:

$$dE = V\,dQ, \quad dE = I\,d\lambda$$

From empty, the voltage rises with the charge and the current rises with the flux, so each store fills the triangle under its line, half the rectangle, the way punch does:

$$E_{can} = \tfrac{1}{2} Q V, \quad E_{coil} = \tfrac{1}{2} \lambda I$$

Every can sentence becomes a coil sentence by trading charge for

flux and voltage for current:

The can	The coil
cans charge, Q	coils linked flux, λ
at a voltage, V	at a current, I
capacitance, C	inductance, L
current is charge's fluxion	voltage is flux's fluxion
cans the capacitance of its	coils the inductance of its
voltage: Q = C V	current: λ = L I
stores the voltage of added	stores the current of added flux:
charge: dE = V dQ	dE = I dλ
holds ½ Q V	holds ½ λ I

Verbing them right:

- A can cans charge, at a voltage. It doesn't can voltage: the voltage is what it's at. "The can cans 24 mC at 12 V."

- A coil coils flux, at a current. It doesn't coil current: the current runs around the turns, and the flux is what threads them. "The coil coils 1.2 Wb at 3.0 A."

- A resistor burns power. It cans and coils nothing.

- A coil is the wire and the store it makes. The wire on a swinging rod is the wire on the rod; the coil is what it stores.

- The rules carry the verbs for the rest: current is the conductance of the voltage, voltage the resistance of the current, power the voltage of the current, and the store the voltage of added charge or the current of added flux.

Two problems, said strong

The can on the battery

> A 12 V battery charges an empty 2000 μF can through a 150 Ω resistor. How much charge does the can can, how fast does the current start, how long does it take, and where does the energy go?

At the start the can is empty and at 0 V, so the resistor takes the whole 12 V. The current is the conductance of 12 V across

150 Ω: 0.080 A.

The can cans charge until it is at the battery's 12 V. It cans the capacitance of 12 V, 2000 µF of 12 V: 24 mC. The resistor and the can together set the time, 150 Ω across 2000 µF, 0.30 s: the can cans about two thirds of its charge in the first 0.30 s and 95 percent of it by 0.90 s.

The battery hands over the voltage of the charge it moves, 12 V of 24 mC: 0.29 J. The can stores the voltage of added charge, and that voltage climbs from 0 to 12 V as the charge comes in, so the can keeps half: the triangle, not the rectangle. The resistor burns the other half. Half and half whatever the resistor: a bigger one takes longer and burns just as much.

The coil that won't let go

> A 0.40 H coil carries a steady 3.0 A. A switch opens, and the coil's only path is through a 10 Ω resistor. How much flux does it coil, what voltage appears, how long does it last, and where does the energy go?

The coil coils the inductance of its 3.0 A, 0.40 H of 3.0 A: 1.2 Wb. The switch opens, and the current can't stop at once. Voltage is flux's fluxion, and dropping 1.2 Wb in no time at all would take a voltage without limit. So the coil keeps the 3.0 A going, through the resistor. The voltage is the resistance of the current, 10 Ω of 3.0 A: 30 V. To make it, the coil lets go of flux at 30 Wb a second.

As the current falls, the voltage falls with it. The coil and the resistor set the time, 0.40 H over 10 Ω, 0.040 s: the same two thirds gone in the first 0.040 s, 95 percent by 0.12 s.

The coil stored the current of added flux, and that current climbed from 0 to 3.0 A as the 1.2 Wb came in: half the rectangle, 1.8 J. The resistor burns all of it. At the start it burns the voltage of the current, 30 V of 3.0 A, 90 W, and the coil's store falls at the same 90 W: the current of 30 Wb a second of flux going out.

Hold the two problems side by side and they're one problem in the mirror. The can took charge in, filling at the voltage; the coil lets flux out, emptying at the current.

The electrical units stay in their own names (V, A, C, F, H, Wb), and the ohm is spelled out wherever the ghost letter Ω is also on the page. The dog park has no volt of its own.

A pattern that slides

A line has a coil and a can in every metre: L_line H and C_line F. Across dt, the voltage dropped along a short dx is the coil's, and the current that goes in at the left and doesn't come out at the right is the can's:

(V_left − V_right) dt = L_line dx dI
(I_left − I_right) dt = C_line dx dV

Ask what it takes for a pattern to slide right at u without changing shape. Across dt the place at x shows what was u dt to its left, so dx dV = u dt (V_left − V_right), and the same for the current. Put that into the two balances. The drops can't both hold unless L_line u across C_line u is 1:

u = 1 on √(L_line across C_line)

and the voltage drop is then L_line u of the current drop. That is the impedance Z, √(L_line on C_line). A pattern sliding right carries I = V on Z; one sliding left, I = −V on Z.

Any state of the line splits that way. V_+ is (V + Z I) over 2, and slides right; V_- is (V − Z I) over 2, and slides left. They put back V = V_+ + V_- and Z I = V_+ − V_-. Each wave moves its own C_line V^2 a metre at u, so its power is V^2 on Z, and nothing steers either one faster than u.

Both waves at +6.0 V on a 50 ohm line: V = 12 V, I = 0. Nothing flows past, and nothing is sitting still: 0.72 W goes right in V_+ and 0.72 W goes left in V_-.

12. Light and clocks (on trial)

Two ships coast apart at a steady pace: A, the active agent, and B, the subject. Each has a clock, and the one thing they can both count is light. A flashes on every tick of A's clock; B sees the flashes arrive. They arrive spaced out by one fixed number, the same on every tick: the shift, s. Going away, the flashes land s times farther apart and the ring is cut by s. Coming at you, they land s times closer and the ring is s times up. By symmetry each gets the same s from the other, and light keeps its own pace for both. That is all it takes. (It is Hermann Bondi's way in Relativity and Common Sense, which builds the whole subject on this one number.)

What the shift gives, with A and B side by side at time 0:

- Pace. A flashes at time t on A's clock. B, going away, sees it at s t on B's clock, bounces it, and A sees the bounce at s s t. The bounce happened halfway between: at A's time (1 + s s) t / 2, and at a distance of light's reach across half the wait, c (s s − 1) t / 2. B's pace is the distance on the time:

  v = c (s s − 1) on (s s + 1)

- Stretch. The same bounce is s t on B's clock and (1 + s s) t / 2 on A's. A's time on B's is the stretch:

  stretch = (s + 1 on s) / 2

  the mean of the shift and its reciprocal. Stretch across the pace on light is half their gap, (s − 1 on s) / 2. At s = 2: pace 3/5 c, stretch 5/4, and 3/4 for stretch across pace. B's clock, going away, reads less than A's by the stretch; lengths along the motion shrink by the same number.

- Pace adds by multiplying. Two hops of 3/5 c are shifts 2 and 2, so s = 4, and the pace is c (16 − 1) on (16 + 1), 15/17 c, 0.88 c. No formula for adding paces is needed.

- Light's energy rides its ring, so it takes the shift. Seen from someone going right at 3/5 c, a pulse running right is cut to half and a pulse running left is doubled: 80 thump becomes 40 and 20 becomes 40. The two sides multiply to 1, which is what it means that the transformation undoes itself.

An event is a place at a time. Frames give an event different places and different times, but they agree on one thing: the time a clock reads when it is carried from one event to the other, its own time. The receiver's own time is 15.6 s, in every frame. S counts 19.5 s, the stretch times more, and the 3.9 s between is lost time, lost only in the counting: from the receiver's side it is S's clock that reads less. In H, where the receiver sits still, nothing is lost.

Check: in S, c across 19.5 s is 5.85×10^9 metres and the receiver goes 3.5×10^9 of them, and 5.85^2 less 3.5^2 is 4.7^2, the same as c across the own time, 15.6 s. In H, the distance is 0.

13. Stealth mode

For dogs who need to pass. Put muscle and skin back on the close-to-the-bone problems. The dog units stay home (SI, with oompf, roll, ponder and wheel), and the vocabulary comes in quietly. Keep the strong language as if it's no big deal: the verb is the argument, slipped past before anyone notices the math is over.

- Start each problem in the textbook's terms, and end each part a little further in. It's "Mary had a little lamb" changing key: the same openings, and the endings reveal the new key.

- Introduce a unit once, in the textbook's terms: "0.072 kg m^2, which we'll call 0.072 wheel". Then just use it: the rod wheels at 15.

- Every unit that comes in gets a verb, an adverb or an adjective: wheel wheels, crank cranks, hook hooks, a bulk ponders from its pivot.

- A few new names per problem, each in its own sentence, never inside the hardest sentence in the room.

- Read it aloud. If you're chewing through "angular acceleration" so often that when another shows up you can still taste the last one, it's time to pick up the pace.

- Introduce the word when the reader is ready to use it. Say the textbook's phrase once in full, and let the sentence start asking for the name, so "we'll call it the pickup" lands as relief. Then it earns its keep at once: the drum picks up, picks up faster, stops picking up.

- Move off sine and cosine and into the arrows by the third problem (section 9).

- Don't point at the ghosts (section 8).

14. Road-legal mode

For dogs on the road in daylight. All the words and none of the costume: the vocabulary is on the table from the first line, the units are the common SI ones, and what a first-year student keeps for good stays put. It is stealth mode without the ramp.

- No ramp. Stealth mode waits until the reader is ready for a word. Here a word comes in when the sentence needs it, with

its definition: one sentence and the equation, then use it. If the sentence needs a paragraph, say it again in the scene's own words, or leave that word out of this problem.

- SI units. Metre, kilogram, second, newton, joule, watt, pascal, m/s, m/s^2, rad, rad/s, N m, kg m^2, kg m/s, coulomb, volt, ampere, farad, henry, weber. No hand, lump, cat, dog, zoom, push, pull, thump, pow, crank, grunt, press, sheet or blanket. Gravity is g = 10 m/s^2, and the rounding rules hold. Oompf, roll, ponder and wheel stay (section 3), each said once against its SI name. A ghost may be counted in cards, 0.01 m^2, where that makes the numbers whole: a profile of 25 card is 0.25 m^2.

- What students keep, stays. Mass, force, weight, velocity, speed, acceleration, momentum, impulse, work, power, kinetic and potential energy, torque, angular velocity, angular acceleration, angular momentum, static and kinetic friction, the normal force, and the laws that go with them: F = ma, p = mv, K = (1/2) m v^2, L = I ω, v = ω r. That knowledge was hard won. Don't swap the nouns for dog nouns; give them the dog verbs: the ball sweeps at 84 rad/s, the friction cranks it back, the centre punches 2.2 J.

- What comes in is the rest: the names for what the usual script leaves as a product or a ratio, and the operators (levered at, across, over, on). Each is defined where it first lands.

Word	Said once	Equation
ponder	mass at a lever	b = m r (kg m)
profile	inertia with the mass out	I / m (m^2)
zing	half the square of the angular velocity	(1/2) ω^2 (1/s^2)
punch	the centre's kinetic energy	(1/2) m v^2
hook	the turning kinetic energy	(1/2) I ω^2
zip	punch per mass	(1/2) v^2 (J/kg)
heft	what gravity stores	m g h
swirl	spin per mass; a point's lever at its velocity	r × v (m^2/s)
ring	the phase rate of anything that repeats	$\dot{\varphi}$ (rad/s)
tone	a swing at one ring	$\ddot{x} = -\text{ring}^2\, x$
shift	the factor flashes arrive farther apart going away	s
stretch	A's time on B's clock	(s + 1/s) / 2

own time	what a clock reads carried	τ
	from one event to the other	
can, coil	the stores of charge and flux	Q = C V, λ = L I

The definitions are the ones in sections 4 to 12, shortened. A definition that reads badly in context gets a closer look before it is used, not a longer sentence.

The same sentence at the three speeds:

Full dog	Road-legal	Stealth, at the start
2 lump with 2 zip punches 4 thump.	The 2 kg centre, at 2 J/kg, punches 4 J.	The 2 kg centre has 4 J of kinetic energy.
40 mw with 200 zing hooks 8.	0.040 kg m^2 with 200 zing hooks 8 J.	The disc has 8 J of rotational energy.

The ball that skids, road-legal

> A uniform solid ball, 1.0 kg and 0.050 m in radius, lies at rest on a level surface. A sharp horizontal blow above the centre sends it off at 2.1 m/s with topspin. Friction, μ = 0.20, then drags it to rolling, at 9/7 of 2.1 m/s. How high was the blow, how long is the skid, and how much energy does it burn?

Profile first: the ball's mass is laid out the same way at any size, and its profile about the centre is (2/5) R^2 = 0.0010 m^2, 0.10 card. With the 1.0 kg back across it, the inertia is 0.0010 kg m^2.

Friction acts at the contact and has no lever about the contact line, and gravity and the surface's push cancel there, so the angular momentum about that line holds from the blow to the roll. Once rolling, the ball turns about its contact, with profile (7/5) R^2 = 0.0035 m^2, sweeping at (2.7 m/s) / R = 54 rad/s: 0.19 roll. Before, the centre's 2.1 oompf levered at R is 0.105 roll, plus the spin, 0.0010 kg m^2 at some ω0. The two agree when ω0 is 84 rad/s, topspin.

The blow's angular impulse about the centre is the spin it

leaves, 0.0010 × 84 = 0.084 roll. The blow itself is 2.1 oompf, so it landed 0.084 on 2.1, 0.040 m above the centre: h = 4R/5.

The skid. The bottom of the ball slides backward at 2.1 m/s - 84 × 0.050 = -2.1 m/s, so friction pulls forward, μ m g = 2.0 N, which speeds the centre up at 2.0 m/s^2: from 2.1 to 2.7 m/s takes 0.30 s, and the centre goes 2.4 × 0.30 = 0.72 m. Check from the other side: 2.0 N at R is a torque of 0.10 N m on 0.0010 kg m^2, 100 rad/s^2, and 84 down to 54 takes 0.30 s.

The energy. Just after the blow the centre punches (1/2)(1.0)(2.1)^2 = 2.2 J, and the spin, at 3500 zing, hooks 0.0010 × 3500 = 3.5 J: 5.7 J. Rolling, it punches 3.6 J and hooks 1.5 J at 1500 zing: 5.1 J. The skid burned 0.63 J, about a ninth. The same from the friction: 2.0 N times the 0.315 m the contact slides, an average of 1.05 m/s for 0.30 s, is 0.63 J.

15. At talking speed

The point of the one-word units is saying this stuff without slowing down. The same sentences, said out loud, from the arm on the post (a 2-lump ball on a 5 hand rod, turned by 2 crank):

Just us dogs	The same, said in full
5 hand sweeping at 6 runs at 3 cat.	A half-metre radius at six radians per second moves at three metres per second.
Half a wheel at 6 has 3 roll.	Half a kilogram metre squared at six radians per second is three kilogram metres squared per second.
2 crank on half a wheel picks it up at 4.	Two newton metres on half a kilogram metre squared gives an angular acceleration of four radians per second squared.
4 pickup over 4.5 stroke adds 18 zing.	At four radians per second squared over four and a half radians, half the square of the angular velocity rises by eighteen per second squared.

Half a wheel with 18 zing	The rotational kinetic energy
hooks 9 thump.	is one half times half a
	kilogram metre squared times
	the square of six radians per
	second, which is nine joules.
+---+---+
3 roll off a 5 hand lever	Three kilogram metres squared
takes 6 oompf.	per second at half a metre is
	six kilogram metres per
	second of linear momentum.
+---+---+
2 grunt through 4.5 stroke	Two newton metres through
adds 9 thump.	four and a half radians is
	nine joules.
+---+---+

The second row says "half a wheel at 6" and drops "sweeping": the row above sets up the parallel, so the ear supplies it. That kind of elision only works where the parallel is that clear. Elision can't be done uniformly, and it's impossible with a tongue covered in algebra.

16. For outside eyes

When doc says a piece is for college, for high school, or for anyone outside the park, the dog units stay in the dog park: SI, with oompf, roll, ponder and wheel wherever those quantities are named. Road-legal mode (section 14) is the usual way to write it. The other rules hold: no third decimal, and name things. The verbs still run:

    The 2 kg centre, running at 3.0 m/s, punches 9.0 J.
    The 2 kg centre, dropping 0.40 m, hefts 8.0 J.

17. A few problems, said strong

Three of the working problems, cut down and said the way the words like to be used.

The drum picks up

    A wheel turns freely on its axle: a uniform 2-lump disc 2 hand in radius, with a uniform 1-lump drum of 1 hand fixed to it on the same axle. A rope
    off the drum runs over a smooth post and down to a 1-lump block. Let go from rest, how long does the block take to fall

a metre?

The profiles go in cards. The disc has half of 4 card, 2 card; the drum half of 1 card. The rope makes the block's fall load the axle like a lump on the drum's rim, so the block presents 1 card at the drum. With the mass back across:

2 lump across 2 card	40 mw
1 lump across half a card	5 mw
1 lump across 1 card	10 mw
the setup	55 mw

The block's bulk on the drum is 1 lump at 1 hand, 0.1 ponder. Under 10 zoom it cranks at 1. 1 crank on the whole 55 mw picks it up at 18, a little over, and 1 hand picking up at 18 speeds the block up at 1.8 zoom. A metre from rest takes a second and a bit, 1.05; the block lands running at 1.9 cat, the wheel sweeping at 19.

The fall lets go 10 thump of heft, and sweeping at 19 adds 180 zing. 55 mw with 180 zing hooks 10 thump: the disc and drum's 45 mw hooks 8.2, the block's 10 mw the other 1.8. The rope carries 10 pull less the 1.8 the block keeps, so 8.2 pull through the metre puts 8.2 thump into the wheel. Check: 8.2 pull at 1 hand cranks the wheel's own 45 mw at 0.82, and that picks it up at 18 too.

The ball that skids

A uniform solid bowling ball, 8 lump and 1 hand in radius, leaves the hand at 7 cat with no sweep at all. While it skids, the lane drags at 16 push. How fast is it going once it rolls?

Set it up about a line across the lane through the launch contact. The drag has no leverage about that line, and the weight's and the floor's leverages cancel, so the spin holds. 8 lump at 1 hand is 0.8 ponder; moving at 7 cat, it has 5.6 roll, backward around. From here on, back around.

Moving the line along the lane to the contact once it rolls adds no leverage on the movement, so the spin stays 5.6 roll. Rolling, the ball turns about its contact: 2/5 card of its own profile and 1 card for the centre, 7/5 card. 8 lump across 7/5 card is 112 mw. 5.6 roll on 112 mw sweeps at 50, and 1 hand sweeping at 50 runs at 5 cat. It keeps 5/7 of its pace.

16 push on 8 lump slows it at 2 zoom, so 7 cat down to 5 takes a second; averaging 6, it skids 6 metres. 8 lump moving at

7 cat punches 196 thump; rolling, 112 mw with 1250 zing hooks 140. The slip falls steadily from 7 cat to nothing, and 16 push through the 3.5 metres it rubs burns the other 56. It travels 6 metres and rubs 3.5.

The puck on a string

A uniform 2-lump puck, 2 hand in radius, lies at rest on smooth ice with string wound on its rim, leaving on the puck's right. A hand steadily pulls at 4, straight ahead along the ice, for a second. How far does the hand go, and how much of its work ends up as punch?

Each ledger gets the whole pull. 4 pull on 2 lump speeds the centre up at 2 zoom. 4 pull at 2 hand cranks at 0.8, and 0.8 crank on 40 mw (2 lump across 2 card) picks it up at 20, forward around. From here on, around.

After the second the centre runs at 2 cat, a metre along; sweeping at 20, the puck has swung through 10 stroke and paid 2 metres of string off the rim. The hand went 3.

As ghosts: 2 zoom over a metre adds 2 zip, and 20 pickup over 10 stroke adds 200 zing. With the bodies back: 2 lump with 2 zip punches 4 thump, and 40 mw with 200 zing hooks 8. 4 pull through 3 metres adds 12 thump, all of it.

Punch's share is the profile over the profile and the radius's reflex: 2 card over 6 card, a third.